

Module 5: NLP Applications & Responsible AI

Text Classification, NER BiLSTM-CRF, BLEU Score, Ethics & Bias

Module V: NLP Applications & Responsible AI — Topics Covered

1. Text Classification & Sentiment Analysis
2. Named Entity Recognition (BiLSTM-CRF)
3. Machine Translation Paradigms & BLEU Score
4. Task-Oriented Dialogue Systems
5. Fairness, Societal Bias & Ethics in NLP

1. Machine Translation Evaluation: BLEU Metric

BLEU (Bilingual Evaluation Understudy) computes modified n -gram precision penalized by brevity:

$$\text{BLEU} = \text{BP} \times \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$
$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r \end{cases}$$

where c is candidate translation length and r is reference corpus length.